

CHAPTER 1

Activities & Lab Packet

Matter, Atoms & the Periodic Table

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 1. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S8P1

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1 Mystery Matter Lab

LAB

OBJECTIVE

Classify samples as element, compound, or mixture.

TIME

30–40 minutes

STANDARDS

S8P1.a, S8P1.c

Materials (per group of 3–4)

- 6 small cups or clear containers labeled A–F
- Samples: sugar (A), salt (B), trail mix (C), iron filings or steel paperclips (D), sand-and-water (E), water (F)
- Magnifying glass (optional)
- Magnet (small)
- Student data sheet (next page)

Setup

Before class, prepare 6 stations or pre-portion samples into cups. Don't tell students what each sample is — let them figure it out from observation. Have the periodic table available so students can verify whether their "element" guesses are real elements.

Student Instructions

1. Examine each sample carefully. Use the magnifier if available.
2. Decide if the sample is one substance or many substances mixed together.
3. If it looks like one substance, decide: is it an element on the periodic table? Or a compound made of two or more elements bonded together?
4. If it's a mixture, decide: can you see the different parts (heterogeneous), or does it look the same all the way through (homogeneous)?
5. Record your answer and the evidence for it on the data sheet.

⚠ SAFETY & TEACHER NOTES

Safety: Tell students not to taste any samples even if they recognize them — common lab rule.

Discussion starter at end: "Sample C (trail mix) and Sample E (sand-water) are both mixtures. Why are they different KINDS of mixtures?"

Common confusion: Students often call salt an "element" because it looks pure. Use this to teach the difference between "pure" (one substance) and "element" (one type of atom).

ANSWER KEY

A — Sugar ($C_{12}H_{22}O_{11}$) = **compound** | B — Salt ($NaCl$) = **compound** | C — Trail mix = **heterogeneous mixture** | D — Iron filings = **element** (Fe) | E — Sand & water = **heterogeneous mixture** | F — Water = **compound** (H_2O)

Mystery Matter — Data Sheet

Name: _____

Period: _____

Sample	What it looks like	My classification	Evidence (why?)
A			
B			
C			
D			
E			
F			

Conclusion: How did you tell mixtures apart from pure substances?

2 Density Tower

LAB

OBJECTIVE

Demonstrate density as a physical property of matter.

TIME

25–35 minutes

STANDARDS

S8P1.c

Materials (per group)

- 1 tall clear cup or graduated cylinder (250 mL works great)
- Honey (about 30 mL)
- Dish soap (about 30 mL)
- Water with food coloring (about 30 mL)
- Vegetable oil (about 30 mL)
- Rubbing alcohol with food coloring (about 30 mL)
- Small items to drop in: paperclip, popcorn kernel, grape, small plastic toy

Setup

Have students predict the order of layers BEFORE pouring. This is the most important part of the lab — predictions force them to engage with density as a concept, not just watch a cool demo.

Student Instructions

1. **Predict first:** Rank the five liquids from densest (bottom) to least dense (top) on your data sheet.
2. Slowly pour honey into the bottom of your cup.
3. Tilt the cup. Slowly pour dish soap down the side. (Pour slowly so it doesn't mix!)
4. Repeat with water, then oil, then alcohol — pouring slowly down the side each time.
5. Drop your small items in one at a time. Observe where each one floats.
6. Record the actual order in your data table. Compare to your prediction.

⚠ TEACHER NOTES

Correct order (bottom to top): Honey → Dish soap → Water → Vegetable oil → Rubbing alcohol

Tip: The "pour slowly down the side" technique is the make-or-break skill here. Demo it first. Some teachers use the back of a spoon to slow the pour.

Extension question: "Why does the paperclip float on dish soap but sink through water?" (Surface tension vs. density — fun trick question.)

Density Tower — Data Sheet

Name: _____

Period: _____

My PREDICTION (bottom to top):

5. (top)

4.

3.

2.

1. (bottom)

ACTUAL results (bottom to top):

5. (top)

4.

3.

2.

1. (bottom)

Where did each object end up?

Paperclip: _____ Popcorn: _____

Grape: _____ Plastic: _____

What does "density" actually mean in your own words?

3 Conservation of Mass

LAB

OBJECTIVE

Show that mass stays the same in a chemical reaction when nothing escapes.

TIME

25–30 minutes

STANDARDS

S8P1.d, S8P1.f

Materials (per group)

- 1 zip-top sandwich bag (must seal tight!)
- 1 small plastic cup that fits inside the bag
- 1 tablespoon of baking soda
- 30 mL of vinegar
- Digital kitchen scale (one per group, or pass between groups)

Student Instructions

1. Put the baking soda inside the small cup.
2. Pour the vinegar into the bottom of the zip-top bag — BUT NOT into the cup yet!
3. Carefully place the cup (with the baking soda) inside the bag. Try not to spill.
4. Press out as much air as possible and seal the bag completely.
5. Place the bag on the scale and record the mass BEFORE the reaction.
6. Pick up the bag and tip the cup over inside it. The vinegar and baking soda will mix and react. You should see bubbles and the bag will inflate!
7. Once the reaction stops, put the bag back on the scale. Record the mass AFTER.
8. Compare the two measurements.

TEACHER NOTES

The big "aha" moment: Mass before should equal mass after. Even though the reaction made gas (CO_2), the gas stayed trapped in the bag — so no atoms left the system.

Extension: Open the bag and let the gas escape. Weigh it again. Now the mass drops! Use this to discuss that atoms aren't destroyed; they just left the system.

Safety: Goggles required. Sealing the bag tight is critical — a loose bag can pop open with surprising force.

Conservation of Mass — Lab Report

Name: _____

Period: _____

My Hypothesis (predict before you start):

I think the mass after the reaction will be: ☐ greater than ☐ less than ☐ the same as the mass before.

Why I think this:

Data:

Measurement	Mass (grams)
Mass BEFORE reaction	
Mass AFTER reaction	
Difference	

Observations during the reaction:

Conclusion: Explain your results using the Law of Conservation of Mass.

4 Build-An-Atom

WORKSHEET

OBJECTIVE

Use the periodic table to identify subatomic particles.

TIME

15–20 minutes

STANDARDS

S8P1.e

TEACHER NOTES

Use this as guided practice or independent work. The "easy trick" from Section 1.7 in the textbook applies: **protons = atomic number**, **electrons = atomic number (in a neutral atom)**, **neutrons = atomic mass (rounded) – atomic number**. Students should have access to a periodic table.

Build-An-Atom Practice

Use a periodic table to fill in the chart. Round atomic mass to the nearest whole number.

Name: _____

Period: _____

Element	Symbol	Atomic #	Atomic Mass	# Protons	# Neutrons	# Electrons
Carbon						
Oxygen						
Sodium						
Iron						
Aluminum						
Helium						
Gold						
Calcium						

Bonus questions:

1. An element has 17 protons. What element is it? _____
2. If a mystery atom has 8 protons and 8 neutrons, what is its atomic mass? _____
3. An atom has 11 protons and 11 electrons. What is its overall charge? _____

ANSWER KEY

Carbon — C, 6, 12, 6, 6, 6 | **Oxygen** — O, 8, 16, 8, 8, 8 | **Sodium** — Na, 11, 23, 11, 12, 11 | **Iron** — Fe, 26, 56, 26, 30, 26 | **Aluminum** — Al, 13, 27, 13, 14, 13 | **Helium** — He, 2, 4, 2, 2, 2 | **Gold** — Au, 79, 197, 79, 118, 79 | **Calcium** — Ca, 20, 40, 20, 20, 20

Bonus: 1) Chlorine (Cl) | 2) 16 | 3) Zero (neutral)

5 Counting Atoms in Formulas

WORKSHEET

OBJECTIVE

Apply subscript and coefficient rules to count atoms.

TIME

15–20 minutes

STANDARDS

S8P1.e, S8P1.f

Counting Atoms Practice

Remember: subscripts (small) apply only to the letter in front. Coefficients (big) apply to EVERYTHING.

Name: _____

Period: _____

Part A: For each formula, count the total atoms of each element.

1. $\text{H}_2\text{O} \rightarrow$ H: _____ O: _____
2. $\text{CO}_2 \rightarrow$ C: _____ O: _____
3. $\text{NaCl} \rightarrow$ Na: _____ Cl: _____
4. $\text{H}_2\text{SO}_4 \rightarrow$ H: _____ S: _____ O: _____
5. $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow$ C: _____ H: _____ O: _____

Part B: Watch out for coefficients!

6. $3 \text{H}_2\text{O} \rightarrow$ H: _____ O: _____
7. $2 \text{CO}_2 \rightarrow$ C: _____ O: _____
8. $4 \text{NH}_3 \rightarrow$ N: _____ H: _____
9. $5 \text{H}_2\text{SO}_4 \rightarrow$ H: _____ S: _____ O: _____

Part C: Parentheses level — the subscript outside applies to EVERYTHING inside.

10. $\text{Ca}(\text{OH})_2 \rightarrow$ Ca: _____ O: _____ H: _____
11. $\text{Al}(\text{NO}_3)_3 \rightarrow$ Al: _____ N: _____ O: _____
12. $2 \text{Mg}(\text{OH})_2 \rightarrow$ Mg: _____ O: _____ H: _____

ANSWER KEY

1) H:2, O:1 | 2) C:1, O:2 | 3) Na:1, Cl:1 | 4) H:2, S:1, O:4 | 5) C:6, H:12, O:6 | 6) H:6, O:3 | 7) C:2, O:4 | 8) N:4, H:12 | 9) H:10, S:5, O:20 | 10) Ca:1, O:2, H:2 | 11) Al:1, N:3, O:9 | 12) Mg:2, O:4, H:4

6 Balanced or Not?

WORKSHEET

OBJECTIVE

Identify balanced chemical equations.

TIME

15 minutes

STANDARDS

S8P1.f

Balanced or Not?

Count atoms on both sides. Circle YES if balanced, NO if not.

Name: _____

Period: _____

#	Equation	Atom counts (left → right)	Balanced?
1	$2 \text{ H}_2 + \text{O}_2 \rightarrow 2 \text{ H}_2\text{O}$		YES / NO
2	$\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$		YES / NO
3	$\text{CH}_4 + 2 \text{ O}_2 \rightarrow \text{CO}_2 + 2 \text{ H}_2\text{O}$		YES / NO
4	$\text{N}_2 + 3 \text{ H}_2 \rightarrow 2 \text{ NH}_3$		YES / NO
5	$\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$		YES / NO
6	$2 \text{ Mg} + \text{O}_2 \rightarrow 2 \text{ MgO}$		YES / NO
7	$\text{H}_2 + \text{Cl}_2 \rightarrow \text{HCl}$		YES / NO
8	$2 \text{ H}_2\text{O} \rightarrow 2 \text{ H}_2 + \text{O}_2$		YES / NO

Bonus: Explain WHY equation #2 is not balanced.

ANSWER KEY

1) YES (H:4=4, O:2=2) | 2) NO (H:4≠2 on right, O:2≠3) | 3) YES (C:1=1, H:4=4, O:4=4) | 4) YES (N:2=2, H:6=6) | 5) NO (Cl:2≠1) | 6) YES (Mg:2=2, O:2=2) | 7) NO (H:2≠1, Cl:2≠1) | 8) YES (H:4=4, O:2=2)

Bonus: H is 4 on left but only 2 on right; O is 2 on left but 3 on right. Atoms don't match.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Star Stuff.** The textbook says you are made of star dust. Explain what this means in your own words. Why is it true?
2. **Phase Change Story.** Pretend you are a water molecule. Tell the story of what happens to you as you go from being in an ice cube → liquid water → water vapor. What changes about how you move?
3. **Compare Two Mixtures.** Pick two of your favorite snacks (one homogeneous mixture and one heterogeneous mixture). Explain how each is mixed and which type it is.

Level 2 — Intermediate (3 paragraphs)

4. **The Detective.** A friend tells you that they "made gold" by mixing two yellow powders together. Use what you know about elements and compounds to explain why this is impossible. What might have actually happened?
5. **Particle Movement.** Describe what is happening to particles in a substance as it heats up from a frozen solid to a boiling liquid. Include what happens during each "flat part" of the heating curve.
6. **Pick an Element.** Choose any element from the periodic table that interests you. Write about its atomic structure, what column it's in, and one cool way it's used in everyday life.

Level 3 — Advanced (essay)

7. **Lavoisier's Law.** The Law of Conservation of Mass changed chemistry forever. Imagine you are explaining it to someone in 1750 who has never heard of atoms. Use evidence from the vinegar-and-baking-soda lab (or another reaction) to convince them that matter is rearranged, not created or destroyed.
8. **Periodic Table Argument.** Some scientists say Dmitri Mendeleev's periodic table was one of the greatest scientific achievements ever. Construct an argument that defends this claim. Use specific examples of patterns the table reveals.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. Is fire matter? Why or why not?
2. Could you have a compound made of just one element? Why or why not?
3. Why is helium on the periodic table but air is not?
4. What happens to the mass of a candle as it burns? Did it disappear?
5. You leave a glass of water in the sun and it disappears. What happened — chemical or physical change?

Deeper Discussions (5–15 min each)

6. **Is salt water still a mixture if you can't see the salt?** How do you know it's still a mixture and not a compound?
7. **Why does mass NOT change in a chemical reaction, even though the products look completely different from the reactants?** Where did the old atoms go?
8. **Imagine an alien scientist visits Earth.** They've never seen the periodic table. How would you explain to them why elements in the same column behave similarly?
9. **The textbook says most of an atom is empty space.** If that's true, why doesn't your hand pass through a desk when you push down on it?
10. **If you broke a sugar cube in half over and over until you reached just one molecule of sugar — and then broke that** — would you still have sugar?

9 Element Trading Cards

GROUP PROJECT

OBJECTIVE

Research and present an element. Build a class-wide reference.

TIME

2–3 class periods

STANDARDS

S8P1.e

Project Description

Each student (or pair) is assigned an element from the periodic table. They create a "trading card" with key facts and an illustration, then present their element to the class. The finished cards are displayed in the classroom to create a class-built periodic table reference.

Card Requirements

Each card (5×7 index card or half-sheet of paper) must include:

- **Front:** Element name, symbol, atomic number, atomic mass, and an illustration (drawn or printed) of one real-world use
- **Back:** Number of protons, neutrons, electrons; what group/family it's in; metal/nonmetal/metalloid; 3 interesting facts; one famous use

Element Assignment Ideas

Easy Elements (good for struggling readers)	Medium Elements	Challenging Elements
Hydrogen, Helium, Carbon, Oxygen, Nitrogen, Aluminum, Gold, Silver, Iron, Copper	Sodium, Calcium, Sulfur, Magnesium, Chlorine, Neon, Potassium, Zinc, Mercury, Lead	Lithium, Argon, Iodine, Tungsten, Uranium, Plutonium, Titanium, Bromine, Krypton, Xenon

Presentation Day

Each student presents for 1–2 minutes. They must include: their element's symbol & number, where it appears on the table, atomic structure, and one fun fact. Other students fill out a "Trading Card Quest" sheet noting one thing they learned about each element.

TEACHER NOTES

Differentiation: Assign elements based on student readiness — easier elements have more accessible information online.

Extension: After the project, post all cards on a large wall periodic table grid. Refer back to them all year.

Grading rubric (suggested): Accuracy of info (40%), creativity (20%), neatness/presentation (20%), oral presentation (20%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S8P1 cross-cutting

Setup

Give each student a blank 5×5 bingo card. Write the vocabulary list on the board. Students fill in their card with vocabulary words (one per square) in any order they choose. The middle square is FREE.

How to Play

Read out a definition or clue. Students cover or mark the matching word on their card. First to get 5 in a row (any direction) wins a small prize.

Vocabulary List (give students this — 24+ words for a 5×5 grid)

matter	atom	element	compound	mixture
homogeneous	heterogeneous	solution	proton	neutron
electron	nucleus	atomic number	atomic mass	periodic table
group	period	metal	nonmetal	metalloid
solid	liquid	gas	plasma	density
physical change	chemical change	reactant	product	balanced equation

Sample Clue Cards (read aloud, don't show)

CLUE TO READ ALOUD	ANSWER
Anything that has mass and takes up space	matter
The smallest part of an element	atom
A pure substance made of just one type of atom	element
Two or more elements chemically bonded together	compound
A mixture you can see the parts of (like trail mix)	heterogeneous
A mixture that looks the same throughout (like saltwater)	homogeneous
The positively charged particle in the nucleus	proton
The negatively charged particle that flies around the atom	electron
The dense center of an atom	nucleus
The number of protons in an element	atomic number
A vertical column on the periodic table	group
A horizontal row on the periodic table	period
State of matter that holds its shape	solid
Superheated gas with charged particles, found in stars	plasma
Mass divided by volume	density
Cutting paper or melting ice is this type of change	physical change
Burning wood is this type of change	chemical change
The substance you START with in a chemical reaction	reactant
What you END with in a chemical reaction	product
When the number of each atom is the same on both sides of an equation	balanced equation

TEACHER NOTES

Easy adaptation: For struggling readers, give them a copy of the vocabulary list with definitions to use during the game — this turns bingo into a low-key matching activity.

Quick prizes: A homework pass, candy, "skip a problem" coupon, or first pick of group partners all work well.

Multiple rounds: Play a few short rounds (regular bingo, X-pattern, blackout) so multiple kids can win.